Adoption of sustainability practices to address environmental concerns was seen as a top priority for majority of the respondents and it also highlighted the need for skills for creating and filling green jobs. The survey findings highlight very strong interest of learners to join this sector.

The survey of trainers highlighted the trend of AI, Green technology, digital and soft skills picking up pace amongst learners as they are seen to have maximum impact on jobs of tomorrow. The survey findings emphasize the support from government required by trainers and institutions in developing industry relevant content and on the job trainings.

Gaining cross-cultural competency and adaptability was additionally seen as an evolving skill and global mobility takes centre stage.

The survey emphasizes the expectation of industry to revamp the basic infrastructure of ITIs and similar institutions by Government so that basic training needs are met leading to enhanced associations between them. It also highlighted the need for improved communication and engagement from skilling institutions which could significantly boost the interest of industry in linking with academic institutions.

This report covered with the inputs of industry's senior leaders and experts complemented with detailed literature review, highlights accelerating transformations in the labour market which will impact the jobs and skills of tomorrow. The report highlights the immediate need for a framework for industry academic linkages in our country. The framework will be able to address the concerns of employers, trainers and learners and recommend ways to navigate through these social, environmental and technological transitions.

Report structure

The report is structured to largely cover three parts. After executive summary, chapters 1 and 2 provide an overview of job and skilling statistics and demographic profile in the labour market briefly mentioning the emerging trends, international best practices and government policy reforms and schemes in the skilling ecosystem.

Chapter 3 deep dives into each sector manufacturing, FMCG, healthcare, energy and infrastructure covering macroeconomic trends, employment scenario, international benchmarking and sectoral trends arrived from primary and secondary research. It concludes the analysis of findings and goes into detail on role of AI/GenAI, DPI perspective, technology, climate change, global mobility, industry 4.0 and 21st century soft skills in the context of skills and employment.

Chapter 4 provides key recommendations for learners, trainers and employers and suggests policy reforms for government to address overall employment challenges and provide supportive role to the industry.

The FOJ 3.0 report is an amalgamation of industry oversight provided by top leaders coupled with detailed literature review of evolving job and skilling landscape in India and the world while highlighting new challenges emerging from disruptive trends and policy measures to redress them.

1

Overview

1.1 Introduction

According to the World Economic Forum's[1] "Future of Jobs Report 2023", the global skills landscape is undergoing significant transformation, driven by the rapid pace of technological change, demographic shifts and evolving job requirements. As per the report, 83 million jobs are projected to be lost but 69 million jobs are projected to be created in the next five years. Technology adoption and broadening of digital access will be the major trends in driving business transformation with big data, cloud computing and AI as the key drivers.

Governments and organizations around the world are investing heavily in upskilling and reskilling initiatives to ensure their workforces remain competitive in the global market. For example, the European Union's Pact for Skills program aims to mobilize a collective effort to upskill and reskill at least 60% of the EU's adult population by 2030[11].

India's skilling and employment landscape is a complex and dynamic ecosystem, influenced by a myriad of factors. Economic policies and technological advancements impact the landscape, apart from demand of skills in an ever-changing employment scenario and the impact of globalization necessitating need for constant change in the skills and employment ecosystem.

This section of the report delves into the demographic dividends, the advent of transformative technologies and the burgeoning gig economy. Additionally, it discusses India's potential to become the skilling capital of the world, the integration of education, skills, and work, and the implications of Generative-Artificial Intelligence, (Gen –AI) Artificial Intelligence (AI), and Digital Public Infrastructure (DPI) on employability, employment and the skilling ecosystem.

Around 1.8 billion people in the world are between the ages of 10 and 24, of which, a sizeable proportion can be found living in developing countries. Projections from United Nations suggest that the youth cohort aged 15-24 is likely to reach around 15.1% of the total population of the world by 2030 and 13.8% by 2050[2]. Presently, among this age group of 15-24, around 500 million survive on less than US$2 per day and more than 73 million face unemployment. Young people, especially in developing countries, face higher socio-economic challenges and are more likely to be employed in the informal sector working in poor conditions. They are either underemployed or unemployed as compared to adults in those countries[4].

> In the Indian context, the large proportion of the young population numbering more than 371 million[5] necessitates the need for innovative and sustainable solutions to convert the demographic advantage of this population into a demographic dividend.

In India, youth face several challenges such as unemployment, gender gaps in the economy, few industry linkages, and low levels of vocational education among others. According to ILO, Annual Report 2023, the unemployment rate in India stands at 4.4% and the Labour Force Participation Rate for women is 33.2% compared to 62.2% for men[6]. Moreover, 57% of women and 34.2% of men in the age group of 15 to 29 years are Neither in Employment, Education or Training (NEET).[7]

The Indian job market is bifurcated into formal and informal sectors. The formal sector encompasses organized employment with regulated working conditions and statutory social security benefits. Conversely, the informal sector is characterized by unorganized employment, with no written contract, paid leave, health benefits or social security. According to ILO, 2024 data, informal sector continues to dominate the Indian labour market, accounting for 71.7% of all workers.

In the midst of an anticipated slowdown in employment, the Indian economy recorded 6.5-7% real GDP in FY25 as per the Economic Survey for 2023-24[8], and Union Budget 2024-2025 announcement of allocating INR1.48 Lakh crore for education, employment and skilling.

In addition, India's demographic profile had a median age of 28.1 in 2023 as per World Population Prospects (WPP) data of UN which is significantly lower than that of many developed nations[9]. This demographic dividend can potentially yield economic prosperity if leveraged through appropriate education and employment opportunities. The challenge lies in equipping this burgeoning young population with the requisite skills to meet the demands of a rapidly evolving job market.

The proliferation of cutting-edge technologies such AIDL, Gen-AI, platform thinking and focus on sustainability, is revolutionizing the industrial landscape of India. At the same time, the country is witnessing a digital transformation and development and adoption of Digital Public Good (DPG) and Digital Public Infrastructure (DPI) across sectors. With this background and buoyed by Government focus on skilling and employment, reforms brought in by National Education Policy 2020 and recent budget announcements, India is on the pathway to be the skilling capital of the world.

> **We are committed to energising skills with a new level of empowerment. In sync with the Industry 4.0 revolution, our focus is on comprehensive skill development, spanning every sector - from agriculture to sanitation. Through 'Skill India' programme, we have ignited growth and a new momentum**
>
> PM Shri Narendra Modi
> *15th August, 2024*[10]

1.2 Trends and disruptions influencing future of jobs

The future of jobs and skills will be shaped by a variety of emerging trends that are driven by technological advancements, industry 4.0 technologies (cloud computing, artificial intelligence, internet of things, robotics and more), environmental concerns, and evolving workforce dynamics. These trends shall not only transform the way we work but also impact the skills required to thrive in the new economy. Here is a look at some of the key trends that are going to impact jobs and skills in the coming decade:

- DPI
- AI and GenAI
- Green technologies
- Workforce Mobility
- 21st century skills

1.2.1 AI and Gen AI

AI might just be the single largest technology revolution of our lifetimes, with the potential to disrupt almost all aspects of human existence. Andrew Ng, Co-founder of Coursera and Professor at Stanford university and formerly founder and Lead of Google Brain, compares the transformational impact of AI to that of electricity 100 years back. With many industries aggressively investing in cognitive and AI solutions, investments in AI are forecasted to approach $200 billion globally by 2025 as per Goldman Sachs report[11].

AI and GenAI are not just impacting existing jobs, they are also creating new ones. The demand for AI specialists, data scientists, machine learning engineers and AI ethics consultants is on the rise. Additionally, the creative capabilities of GenAI are leading to new roles in content creation, digital marketing and design, where AI-generated content can be used as a starting point or a tool for human creativity. This shift underscores the need for a workforce that is skilled in these new technologies, capable of leveraging AI to enhance their work rather than being replaced by it.

The report explores the impact of AI in the five key sectors for this study, i.e., manufacturing, healthcare, infrastructure, energy and FMCG.

- In manufacturing, AI optimizes production lines and reduces downtime.
- In healthcare, AI enhances diagnostics and personalized treatment plans.
- The infrastructure sector benefits from AI in project management and smart city development.
- In the energy sector, AI helps optimize energy consumption and integrate renewable sources.
- In FMCG, AI improves supply chain management and customer insights. These advancements demonstrate the transformative potential of AI in driving efficiency and innovation across India's diverse economic landscape.

The sectors expected to benefit most include business services, finance, transportation, education, retail and healthcare due to their digitalization and focus on productivity, efficiency and personalized experiences.

1.2.2 Going green

The global push towards sustainability and the reduction of carbon footprints is influencing job markets. The 'green economy' encompasses renewable energy, sustainable transportation, energy efficiency, waste management and green construction. This shift towards sustainability is driven by the need to combat climate change, conserve resources and promote environmental responsibility. Jobs in these sectors are expected to grow as countries invest in cleaner technologies and infrastructure. As a result, there is a growing demand for green jobs—roles that contribute to preserving or restoring the environment, whether in traditional sectors like manufacturing or emerging fields like renewable energy. Workers with expertise in environmental science, sustainable design, green technology and environmental regulations, audit compliances will be in high demand. According to Global Green Skills Report 2023, the share of green talent in the workforce rose by a median of 12.3% across the 48 countries between 2022 and 2023, whereas the share of job postings requiring at least one green skill grew by a median of 22.4%.[12]

In India, the focus on going green is having a notable impact on five key sectors identified for this study.

In manufacturing, sustainable practices are being integrated into production processes to reduce waste and energy consumption.

The healthcare sector is adopting green initiatives to minimize environmental impact through eco-friendly hospital designs and waste management systems.

Infrastructure projects are increasingly incorporating sustainable materials and energy-efficient designs.

The energy sector is at the forefront of this movement, with a significant shift towards renewable energy sources such as solar and wind power.

In FMCG, companies are prioritizing sustainable sourcing and packaging to meet consumer demand for eco-friendly products. According to the Green Industry Outlook report [16] by Team Lease Digital, India has the potential to double the current employment of 18.52 million green jobs to 35 million by 2047, with contractual workers being major contributors to the industry's growth.

1.2.3 21st century skills

21st century skills refer to the knowledge, life skills, career skills, habits and traits that are critically important to a student's success in today's world, particularly as they move on to college, the workforce and adult lives. The demand for 21st-century skills is transforming the future of jobs, emphasizing the need for competencies that align with the evolving digital and global landscape. These skills include critical thinking, creativity, collaboration, communication, digital literacy and adaptability.

The top 10 future skills as per World Economic Forum are:

- Analytical thinking and innovation
- Active learning and learning strategies
- Complex problem-solving
- Critical thinking and analysis
- Creativity, originality and initiative
- Leadership and social influence
- Technology use, monitoring and control
- Technology design and programming
- Resilience, stress tolerance and flexibility
- Reasoning, problem solving and ideation

As technology continues to advance rapidly, traditional job roles are being redefined, necessitating a workforce capable of navigating complex problems, innovating solutions, and working effectively in diverse and dynamic environments. Educational institutions and training programs are increasingly focusing on imparting these essential skills to prepare individuals for the future job market.

1.2.4 Digital Public Goods (DPG)/Digital Public Infrastructure (DPI)

DPGs are defined by the Digital Public Goods Alliance (DPGA) as "open-source software", open data, open artificial intelligence models, open standards and open content. As of 2024, there are 162 DPGs in existence[13]. Built on free, open-source software, the technology allows governments and organizations to take existing solutions, customize and adapt them, and integrate them into their own digital infrastructure, greatly expediting digital transformation around the world. In the coming years, DPGs will be critical to digital transformation as they offer cost-effective, tried and tested solutions that allow the digital transformation of citizen-focused services and are aligned with SDGs. The DPG market is still young, but growing fast, and expected that it will be worth around US$100b by 2030[14].

Digital Public Infrastructure - DPI refers to fundamental digital solutions that enable and catalyze the provision of society-wide functions and services, both in the public and private sectors. They are typically population-scale systems on which the digital public services operate, such as identity systems (for example, India's Aadhaar system), payment systems (such as Unified Payment Interface (UPI)) and data exchange networks (for example, Estonia's X-Road). DPI enables effective provision of essential society-wide functions and services in the public and private sectors. A country's digital public infrastructure may include implementations of either multiple proprietary or open-source solutions (including DPGs) or both.

> DPI plays a crucial role in driving economic growth, improving service delivery, and enhancing citizen engagement. By providing a solid foundation for digital services, DPI can enable countries to achieve their development goals and improve the lives of their citizens.

A leading example of DPI in skilling, education, employment and entrepreneurship is Skill India Digital Hub (SIDH)[15]. It is built on principles of open source, inter operable and scalable framework. Launched in September 2023 by Hon'ble minister Shri Dharmendra Pradhan, it was also declared as 'the Digital Public Infrastructure (DPI) for skilling, education, employment and entrepreneurship ecosystem of India .' Some of the unique features of the platform are digital verified credentials (DVC) for all learners, recommendations for learning and job matching, integration with other government schemes and a range industry relevant courses. Currently it has over 9 million registrations and 750+ self-paced courses in 11 languages.

1.2.5 Workforce mobility

India's vision of becoming the skilling capital of the world is driven by its demographic advantage and the government's strong emphasis on skill development. With initiatives like the National Skill Development Mission and the Skill India program, the country has embarked on an ambitious path to train millions of its citizens in a wide range of sectors, from traditional trades to emerging technologies. The focus is on creating a workforce that is not only skilled but also globally competitive.

The global demand for skilled Indian labour is robust, with countries in the Middle East, East Asia, Europe, and North America being key destinations. Enhancing workforce mobility through international accords and the global recognition of Indian qualifications is crucial in capitalizing on this demand.

International mobility

The international demand for skilled labour is on the rise, particularly in sectors such as information technology, healthcare, engineering and construction. India's skilled workforce is highly sought after in these industries due to their technical expertise, proficiency in English, and adaptability to different working environments. Countries in the Middle East, Europe, North America and Asia-Pacific are increasingly looking to India to fill gaps in their labor markets, especially in fields where there is a shortage of local talent.

As per the recent NSDCI report[16] of 16 high potential countries (Saudi Arabia, UAE, USA, Canada, Qatar, Kuwait, Oman, Bahrain, Australia, Germany, Japan, UK, Singapore, Malaysia, Sweden and Romania) to understand India's overseas employment landscape,

- A five-year potential of approximately 3.9 million Indian workforce has been identified for international deployment in these countries.
- Sectors like construction, healthcare – personal care and social care, hospitality and tourism management, energy (oil & gas and renewables), education (teachers), shipping and logistics, IT and Digital, retail, manufacturing and media & entertainment.
- A few of the sector observations and demand insights are as follows:

IT and technology sector: India has long been recognized as a global leader in information technology (IT) and software services. The country's IT professionals are known for their technical skills, problem-solving abilities, and experience with cutting-edge technologies such as Artificial Intelligence (AI), Machine Learning, and Blockchain. As digital transformation accelerates across industries worldwide, the demand for Indian IT professionals continues to grow, with countries like the United States, Canada, and the United Kingdom being key destinations for Indian tech talent.

Healthcare and allied services: The global healthcare sector is also experiencing a surge in demand for skilled professionals, particularly in nursing, caregiving and medical technology. Indian healthcare workers are valued for their expertise, cultural sensitivity, and ability to work in high-pressure environments. Countries facing aging populations, such as Japan and Germany, are actively recruiting Indian healthcare professionals to address their workforce shortages. The recognition of Indian medical qualifications by several international bodies further boosts the mobility of Indian healthcare workers.

Engineering and construction: India's engineers and construction workers are known for their technical proficiency and hands-on experience in large-scale infrastructure projects. The Middle East has been a significant destination for Indian engineers and construction workers, driven by the region's ambitious infrastructure development plans. Indian professionals in these fields are contributing to the construction of mega-projects, smart cities, and renewable energy initiatives across the globe.

Some of the sunrise sectors[17] providing growth impetus are-

- **Electronics and semiconductors:** India's electronics industry is poised for substantial growth, with projections indicating a production value of $300 b by FY26. The semiconductor market is estimated to touch $64 b by 2026, almost three times its 2019 size of $22.7 b. This surge is fuelled by initiatives such as the Production Linked Incentive (PLI) scheme, aimed at bolstering large-scale electronics manufacturing under the Make in India campaign. With domestic production accounting for 65% of the electronics market valued at $155 b, India is emerging as a key player in the global electronics landscape.
- **Electric vehicles (EV):** India has set an ambitious target of having 30% of all vehicles electric-powered by 2030. To facilitate this, the government has implemented a multifaceted approach. It has opened doors to 100% foreign direct investment (FDI) in the EV sector, encouraging international players to invest in India's burgeoning electric mobility landscape. Furthermore, the establishment of over 12,146 operational public EV charging stations nationwide, coupled with schemes like FAME II (Faster Adoption and Manufacturing of Electric Vehicles) which includes financial support in the form of subsidy for setting up Public Charging Infrastructure to instil confidence among the EV users, underscores the government's commitment to incentivizing EV adoption across various vehicle segments.
- **Renewable energy:** With the third largest energy consumption globally, India's transition to renewable energy is pivotal for global climate action. The country's enhanced target of achieving 500 GW of non-fossil fuel-based energy by 2030 marks the world's largest expansion plan in renewable energy.
- **Agro and food processing:** The agricultural sector remains one of the key drivers of the Indian economy. Significant drivers fuelling this growth include increasing demand, expanding exports, supply-side advancements such as hybrid seeds and advanced irrigation infrastructure, and supportive government policies.

Other sectors include, Artificial Intelligence, Geospatial Systems and Drones, Space Economy, Genomics and Pharma, Green Energy and Clean Mobility Systems.

Some of the key trends and disruptions impacting World of Work (WoW)-

- **Gig economy:** The gig economy which refers to short-term work contracts or freelance work, has gained significant traction in India, offering flexible employment opportunities through platforms. Youth participation in the gig economy has increased 8-fold between 2019 and 2022. The gigification (which also includes platformization) of jobs and the acceptance of remote work are two important trends that have appeared in the labor market in recent years.
- **Reducing industry – skill gap:** Another trend is how educational institutions are increasingly aligning with industry requirements, ensuring that academic curricula are attuned to the practical needs of the job market. The proliferation of vocational training and apprenticeship programs is bridging the gap between theoretical knowledge and practical application, thus fortifying the workforce's employability.
- **Platform thinking:** As India is poised to become a global epicentre for skill development, the 'Skill India Digital Hub (SIDH)' mission is a testament to the nation's commitment to train millions in a plethora of vocational trades.
- **Integration of WoE-WoS-WoW:** The disruptions brought about by globalization and technological advancements have necessitated a seamless integration of the Worlds of Education (WoE) – World of Skills (WoS) – World of Work (WoW).

  Key trends driving this integration include:

  - **Lifelong learning:** As the pace of technological change accelerates, individuals need to continuously update their skills to remain competitive. Lifelong learning initiatives are becoming increasingly important in ensuring that the workforce has the necessary competencies.
  - **Digital transformation:** The digital revolution is transforming the way we work and learn. Online education platforms, digital tools and virtual work environments are becoming increasingly prevalent, blurring the boundaries between education, skills and work.

In summation, the employment and skilling ecosystem in India is at a pivotal juncture, with demographic advantages, technological advancements, and globalization shaping its trajectory. The nation's capacity to harness its demographic potential, embrace technological innovation, and cultivate an inclusive workforce will be determinative of its role as a global leader in skills. The harmonization of education, skills and work, coupled with proactive policy measures to adopt the future skills will be critical in navigating the challenges and capitalizing on the opportunities that the new world of work will bring.

2

India skilling landscape